

# Alex Siegel

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## SUMMARY

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I'm a research scientist with over 10 years of experience applying biophysical and biochemical techniques to study the mechanisms of proteins. My recent work provided the detailed mechanistic understanding of how protein chaperones preserve their misfolding clients and offers a platform to build custom chaperones that combat aggregation-related diseases. I want to apply my expertise to study the molecular mechanisms of proteins associated with diseases and contribute to the development of drugs and other novel therapeutics.

## RESEARCH EXPERIENCE

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\*Indicates equal contribution first author publications.

### Postdoctoral Scholar in Chemistry at Caltech

Research with Prof. Shu-ou Shan

Pasadena, CA  
2017-present

- I developed a pipeline to map transient protein-protein interactions between a chaperone (cpSRP43) and its clients by crosslinking-coupled mass spectrometry ([McAvoy, Siegel, et al, J. Biol. Chem, 2018](#)).
- We identified a new class of cpSRP43 clients involved in chlorophyll synthesis ([Wang et al, Proc. Nat. Acad. Sci., 2018](#)) and I discovered cpSRP43's heat stress protection towards them ([Ji\\*, Siegel\\* et al, Nature Plants, 2021](#)). This work may provide a new target for genetic engineering of more heat-tolerant plants.
- I discovered a novel activation mechanism of cpSRP43 triggered by an order-to-disorder transition ([Siegel\\*, McAvoy\\* et al, J. Mol. Biol., 2020](#)), and showed how its conformational dynamics targets it to different protein clients ([Siegel et al, In Revision at Science Advances, 2025](#)) or to non-native amyloid-forming clients ([Prov. Patent ID: CIT-9049-P](#)). This work provides a bioengineering platform to tailor chaperones against toxic aggregates found in neurodegenerative diseases.
- I solved the cryo-EM structure of human mitochondrial ClpB homolog, Skd3, revealing its roles as both a classic disaggregase chaperone and a unique chaperonin-like folding cage ([Gupta\\*, Lentzsch\\*, Siegel\\*, et al, Science Advances, 2023](#)). This work enables the design of new drugs towards several congenital mitochondrial diseases.

### Graduate Researcher in Biophysics at UC Berkeley

Research with Prof. David Wemmer

Berkeley, CA  
2009-2016

- I uncovered how bacterial transcriptional activators recognize and thread a subunit of RNA polymerase to apply the force needed to melt DNA and initiate transcription of mRNA ([Siegel et al, J. Mol. Biol., 2016](#)).
- I used molecular tweezers to understand the force-dependent activation of RNA polymerase ([Siegel, dissertation](#)).

## EDUCATION

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### University of California, Berkeley

Ph.D. in Biophysics with specialization in Chemical Biology  
Graduate advisor – Prof. David E. Wemmer, Chemistry  
Thesis: Mechanisms of  $\sigma^{54}$  bacterial transcription activation

Berkeley, CA  
2016

### California Institute of Technology

B.S. in Biology and Chemistry  
Double major, graduated with Honors

Pasadena, CA  
2008

## SKILLS

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\*Experimental or computational skills with demonstrated proficiency (instruments/software/methods used)

- I'm an independent researcher able to initiate original projects outside my expertise from gaps in primary literature.
- I'm an experienced science communicator, presenting at multiple large conferences and drafting 9 published papers.
- I've mentored and directly trained 20 student researchers, including as a project leader for group experiments.
- I have lengthy experience with a broad range of techniques in biophysics, biochemistry, and molecular biology (see below), and I learn new methods quickly.

**NMR Spectroscopy.** 15 years of experience studying protein structure and dynamics by NMR Spectroscopy

- Isotope labeling (M9 media growth)
- Spectra collection (Bruker Topspin)
- Writing pulse programs (Bruker Topspin)
- Data analysis (NMRpipe, NMRview, CARA, MestreNova)
- Spectrometer maintenance (Bruker 600-900 MHz, nitrogen & helium fills, basic maintenance).
- Site-specific labeling (MTSL & MMTS, <sup>19</sup>F-Phe, BTFA)
- Multi-dimensional collection and analysis (Diffusion ordered spectroscopy, triple resonance assignments, structure determination by NOE)
- Paramagnetic relaxation enhancement
- Non-uniform sampling

**Biophysics Techniques.** Biophysical techniques to characterize protein structure, dynamics, and interactions

- X-Ray Crystallography
- Mass Spectrometry (Orbitrap Elite, Q-Exactive)
- Electron Paramagnetic Resonance (Bruker CW-EPR)
- Cryo Electron Microscopy (Titan Krios)
- Small Angle X-ray Scattering
- Fluorescence Anisotropy and FRET (Horiba Fluorolog)
- Isothermal titration calorimetry (MicroCal iTC200)
- Microscale thermophoresis (Monolith NT.115)
- Single molecule FRET (ALEX, Hamamatsu Photonics picosecond streak camera)
- Fluorescence lifetime measurements (Horiba TCSPC)
- Mass Photometry (Refeyn)
- Circular Dichroism (Aviv model 410)
- Molecular Tweezers

**Molecular Biology Techniques.** Cloning, purifying, and *in vitro* assays to study proteins.

- Cloning (site-directed mutagenesis, Gibson Assembly)
- Protein expression (E. coli)
- PCR/qPCR
- Protein purification (FPLC/Chromatography)
- Immunoassays (western blotting, immunoprecipitation)
- Amber suppression (Schultz method)
- In vitro translation (PURE and self-prepared systems)

**Biochemistry Techniques.** Labeling and other biochemical assays to monitor protein activity

- Light scattering (Beckman DU series)
- Luciferase folding assays (SpectraMax iD5 plate reader)
- ATPase assays
- Chemical crosslinking (BMOE, BMH, DSSO, DHSO)
- Fluorescence labeling (AlexaFluor/BODIPY maleimides)
- EPR and NMR labeling (MTSL)
- Chromatography (Agilent 1100 series HPLC).

**Software.** Experience with various protein structure packages, bioinformatics tools, and coding languages

- Structure visualization (ChimeraX, Pymol)
- Cryo-EM analysis (CryoSPARC, Relion)
- Protein structure determination (PHENIX, Coot)
- Mass spectrometry (ProteinProspector, MeroX)
- Structure computation (Gromacs, AlphaFold 2)
- Bioinformatics (BioPython)
- Data analysis (R, Matlab, Graphpad Prism)
- Coding languages (Python)

## SERVICE

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- **Research Mentorship** 2010-present  
I trained 20 student scientists in biochemistry, molecular biology, and biophysics projects, including: 1 postdoc, 7 PhD students, 6 undergraduates, and 6 high school volunteers. These led to several co-authored papers and presentations.
- **Caltech Postdoctoral Association (CPA)** 2018-2022  
I served as Secretary and Vice Chair and worked to improve the postdoctoral experience at Caltech. We successfully lobbied for increased event and career development funding, planned dozens of career and social events including the Los Angeles wide ExploreCaltech outreach event, improved onboarding of new postdocs, and coordinated postdoc interests across different academic divisions.
- **UC Berkeley Graduate Assembly Delegate (GA)** 2011-2012  
I represented the Biophysics Graduate Group at UC Berkeley and advocated department issues on behalf of other graduate students.
- **Caltech Academic Research Council (ARC)** 2005-2007  
I served as a representative and Secretary of the ARC. We gathered course feedback and liaised with faculty to propose and implement specific improvements to undergraduate curriculum and teaching.